

Low-Noise 5V DC CONVERTER Using LM2574

PETRE TZV PETROV

This circuit is used to convert DC voltages like 9V and 12V to 5V, 500mA DC, which is suitable for embedded systems.

Use of a linear regulator normally found in an AC-DC adaptor is not a good idea, because it requires a large heat-sink to dissipate thermal power. Presented here is a simple, low-noise and low-cost solution for a DC-DC converter using LM2574.

Circuit and working

The circuit diagram of the low-noise 5V DC converter with LM2574 is shown in Fig. 1. It is built around step-down voltage regulator LM2574-5.0 (IC1), three LEDs (LED1 through LED3), rectifier diode 1N4007 (D1),

Schottky diode 1N5819 (D2) and a few other components.

The circuit employs an input filter and a diode for reverse voltage protection. LM2574 is available with fixed outputs like 3.3V, 5V, 12V and 15V, and adjustable versions from 1.23V to 37V. The IC is identified by the last digit(s). For example, LM2574-5.0 means it is a fixed 5V output IC.

Input voltage across CON1 can be regulated or unregulated from 7V to 40V. Actual input depends on voltage ratings of input capacitors and rectifier diodes D1 and D2. Here, the circuit is designed for 12V DC input.

LED1 and LED2 show the presence of input voltages with polarity indication. That is, LED1 is on if polarity

of input voltage is wrong (negative). LED2 is on when polarity of input voltage is correct (positive).

Fuse F1 is resettable and selected according to the current capabilities of the voltage source and protection diode D1. You may replace F1 with a simple 0.5A fuse for single use and D1 with an appropriate transil. D1 can be 1N540X or 1N1400X, and is selected according to maximum input voltage and F1 rating. Resistor R2 (0.1-ohm, 0.5W) is used to monitor input current of the converter.

PARTS LIST

Semiconductors:

- IC1 - LM2574-5.0, step-down 5V voltage regulator
- D1 - 1N4007 rectifier diode
- D2 - 1N5819 Schottky diode
- LED1-LED3 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 1.5-kilo-ohm
- R2 - 0.1-ohm, 0.5W
- R3 - 1-kilo-ohm

Capacitors:

- C1 - 1nF ceramic disk
- C2, C5, C6 - 100nF ceramic disk
- C3, C4 - 100 μ F, 63V electrolytic
- C7 - 220 μ F, 16V electrolytic
- C8 - 100 μ F, 16V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- F1 - 1A fuse with holder
- L1, L2 - 10 μ H inductor
- L3, L4 - 330 μ H inductor
- 12V power supply

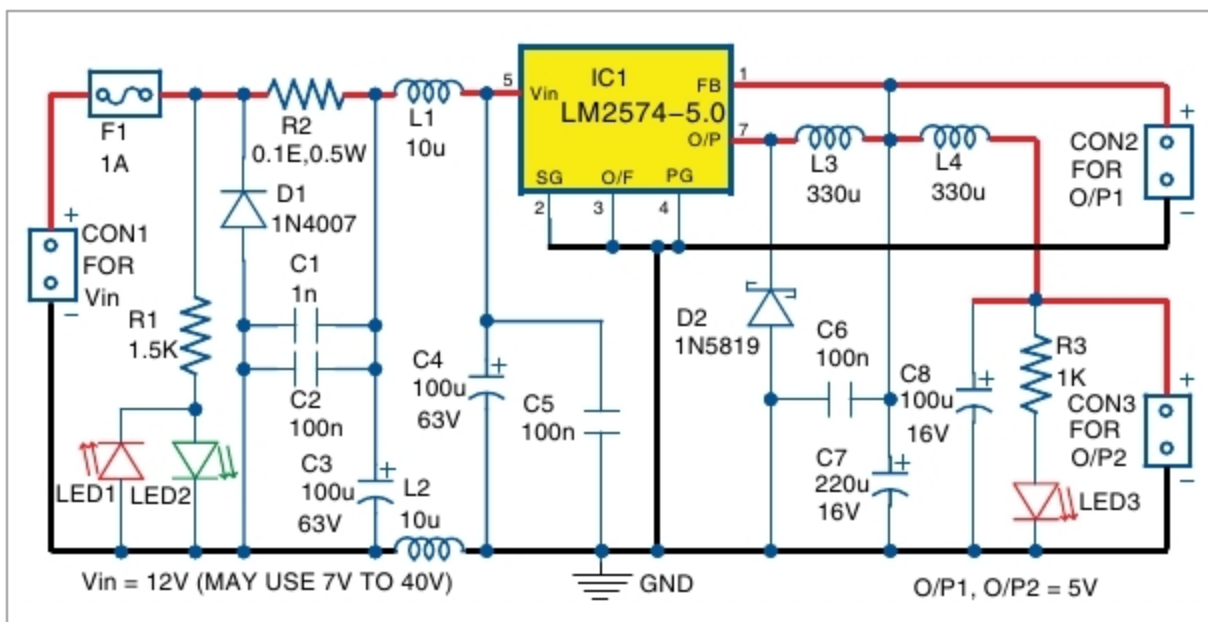


Fig. 1: Circuit diagram of 5V DC converter

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou

The latest from the Open Source world is here.

OpenSourceForU.com

Join the community at facebook.com/opensourceforu

Follow us on Twitter @OpenSourceForU